

SUMMER INTERNSHIP PROJECT REPORT

Road Accident Severity Analysis and Prediction using Machine Learning

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1. Introduction

Road accidents remain one of the most pressing public safety concerns worldwide, claiming approximately 1.35 million lives each year according to the World Health Organization. Beyond the tragic loss of life, road accidents impose a substantial economic burden on healthcare systems, insurance providers, and national economies. In India alone, road accidents account for nearly 150,000 deaths annually, making it one of the highest accident-prone nations in the world.

Understanding the factors that contribute to accident severity -- whether an accident results in a minor injury, a serious injury, or a fatality -- is critical for designing effective road safety policies. Traditional accident analysis methods, which rely on manual investigation and basic statistical summaries, often fail to capture the complex, non-linear relationships between multiple contributing factors such as weather conditions, road surface type, driver demographics, vehicle characteristics, and time of occurrence.

Machine Learning (ML) offers a powerful alternative by enabling algorithms to automatically learn patterns from historical accident data and make predictions about future incidents. By leveraging techniques such as ensemble learning, feature engineering, and class imbalance correction, it is possible to build predictive models that can identify high-risk scenarios and support data-driven decision-making in road safety management.

This project presents a comprehensive approach to road accident severity analysis and prediction. Using a real-world dataset of over 12,000 accident records, we developed and evaluated multiple machine learning models -- including Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, and CatBoost -- to predict whether an accident will result in a Slight Injury, Serious Injury, or Fatality. The best-performing model, an Enhanced XGBoost classifier trained on 738 engineered features with ADASYN oversampling, achieved an accuracy of approximately 93% on the full dataset. The trained model has been deployed as an interactive Streamlit web application that allows users to explore accident statistics and obtain real-time severity predictions.

2. Motivation

Every year, road accidents claim thousands of lives and leave many more with life-changing injuries. Despite continuous improvements in road infrastructure and traffic management, accidents continue to occur due to a combination of factors -- adverse weather conditions, poor visibility, inexperienced drivers, vehicle defects, and hazardous road layouts. The human and economic cost of these incidents makes road safety a matter of urgent national importance.

Traditional approaches to accident analysis typically involve manual investigation by traffic authorities, which is both time-consuming and limited in scope. These methods often fail to uncover the hidden patterns that exist within large, multi-dimensional accident datasets. As a result, the insights gained are frequently reactive rather than proactive -- responding to accidents after they happen rather than predicting and preventing them.

This project is motivated by the belief that data-driven approaches can fundamentally change how we understand and respond to road accidents. By applying machine learning algorithms to historical accident data, we can identify the key factors that influence severity, predict the likely outcome of future accidents, and provide actionable insights to traffic authorities, urban planners, and emergency response teams.

Furthermore, the project demonstrates that advanced analytical tools are not limited to large corporations or research institutions. With open-source technologies like Python, Scikit-learn, and Streamlit, it is possible to build powerful predictive systems that are accessible, scalable, and ready for real-world deployment. This project serves as a practical demonstration of how machine learning can be applied to solve critical problems in transportation safety and public health.

3. Objectives

The primary objectives of this project are:

- To perform a thorough exploratory data analysis (EDA) of road accident data in order to identify the key factors -- such as weather, road conditions, driver demographics, and time of day -- that influence accident severity.
- To clean, preprocess, and transform the raw dataset by handling missing values, encoding categorical variables, engineering meaningful features, and addressing class imbalance through oversampling techniques.
- To develop and evaluate multiple machine learning classification models -- including Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost -- for predicting accident severity.
- To compare the performance of baseline models against hyperparameter-optimized versions using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.
- To identify the best-performing model and ensure its consistency between the training notebook and the deployed application.
- To build an interactive Streamlit-based web dashboard that enables users to explore accident statistics, visualize trends, and obtain real-time severity predictions using the trained model.
- To provide actionable insights that can support road safety policy decisions and contribute to reducing the incidence and severity of road accidents.

4. Scope

This project encompasses the complete machine learning pipeline -- from data acquisition and exploratory analysis through model development, evaluation, and deployment.

- **Dataset:** The project uses the "Road.csv" dataset containing 12,316 accident records across 32 attributes, covering weather conditions, road surface type, light conditions, driver demographics, vehicle information, and accident severity labels.
- **Feature Engineering:** Over 700 features are derived from the original 32 columns through temporal feature extraction, domain-specific risk scoring, binary interaction flags, context combination features, frequency encoding, and one-hot encoding.
- **Model Development:** Six machine learning algorithms are implemented and compared, with hyperparameter optimization using RandomizedSearchCV. The final deployed model is an Enhanced XGBoost classifier trained with ADASYN oversampling.
- **Deployment:** The trained model is integrated into a Streamlit web application featuring interactive data visualization, real-time prediction, feature importance analysis, and model comparison dashboards.
- **Evaluation:** Model performance is assessed using standard classification metrics including accuracy, precision, recall, F1-score, confusion matrices, and per-class analysis.

5. Dataset Description

The dataset used in this project is the "Road.csv" file, which contains records of road accidents collected from official traffic authority databases. The dataset comprises 12,316 individual accident records, each described by 32 attributes covering a wide range of environmental, vehicle, and human factors.

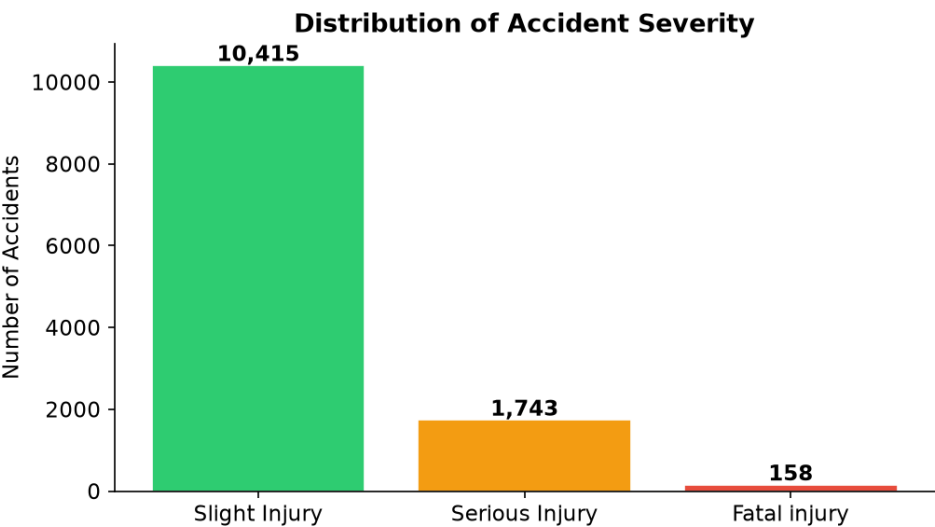
5.1 Dataset Overview

Attribute	Value
Total Records	12,316
Total Features	32
Target Variable	Accident_severity (3 classes)
Missing Values	Present in 16 columns (max 35.9%)
Duplicate Records	0

5.2 Target Variable Distribution

The target variable, Accident_severity, is a categorical feature with three classes. The dataset exhibits significant class imbalance, which is a common characteristic of real-world accident data.

Severity Class	Count	Percentage
Slight Injury	10,415	84.56%
Serious Injury	1,743	14.15%
Fatal injury	158	1.28%



5.3 Key Feature Distributions

Understanding the distribution of key features provides valuable context for the analysis.

Weather Conditions:

Weather	Count	%
Normal	10,063	81.71%
Raining	1,331	10.81%
Other	296	2.40%
Cloudy	125	1.01%
Windy	98	0.80%
Snow	61	0.50%

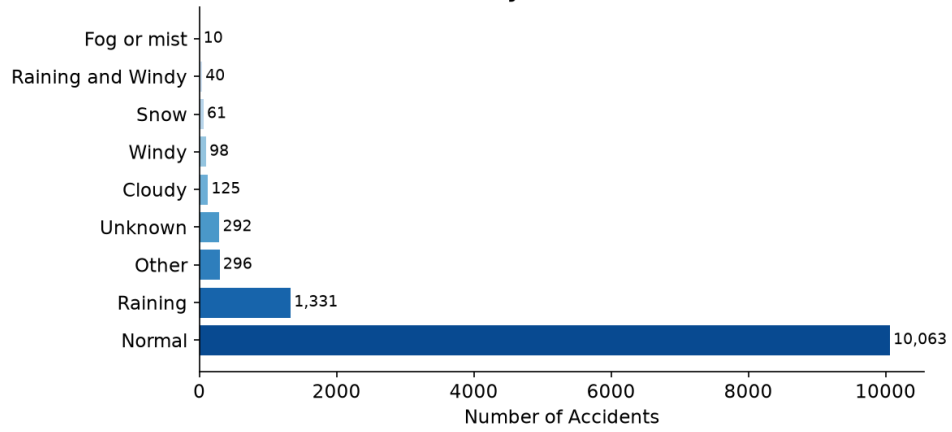
Light Conditions:

Light Condition	Count	%
Daylight	8,798	71.43%
Darkness - lights lit	3,286	26.68%
Darkness - no lighting	192	1.56%
Darkness - lights unlit	40	0.32%

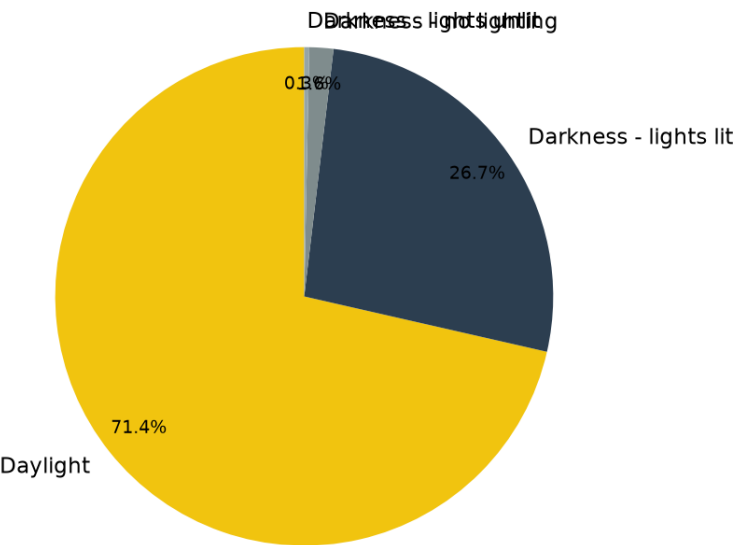
Driver Age Distribution:

Age Band	Count	%
18-30	4,271	34.68%
31-50	4,087	33.19%
Over 51	1,585	12.87%
Unknown	1,548	12.57%
Under 18	825	6.70%

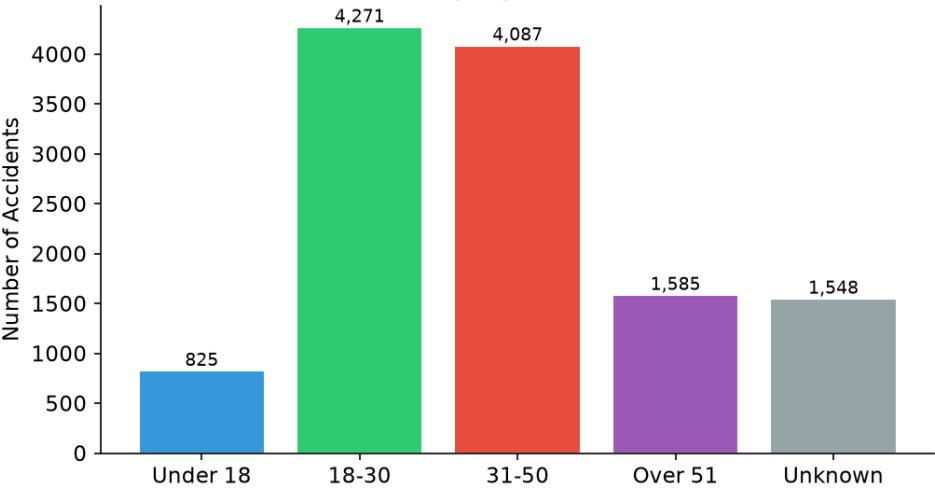
Accidents by Weather Condition

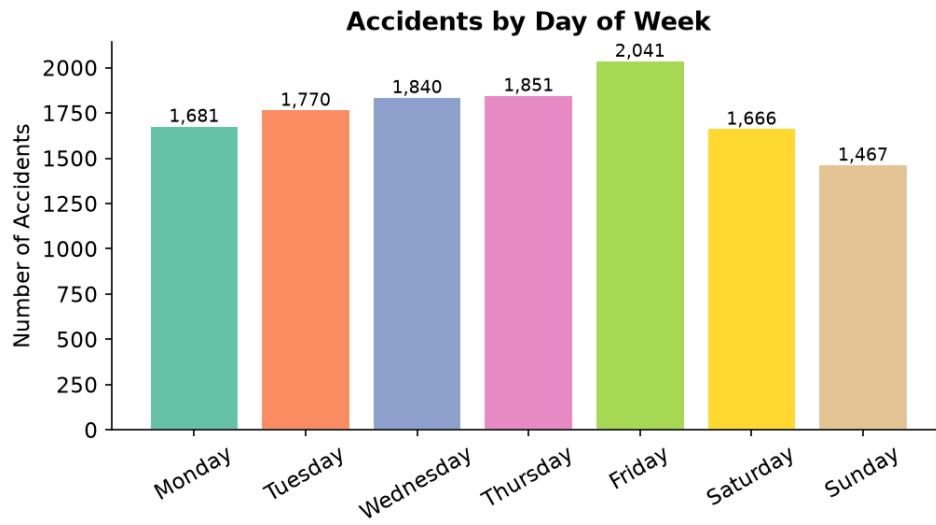
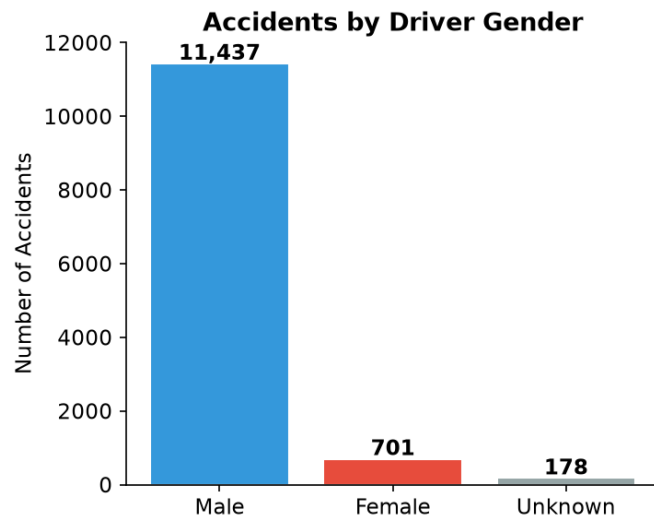


Accidents by Light Conditions

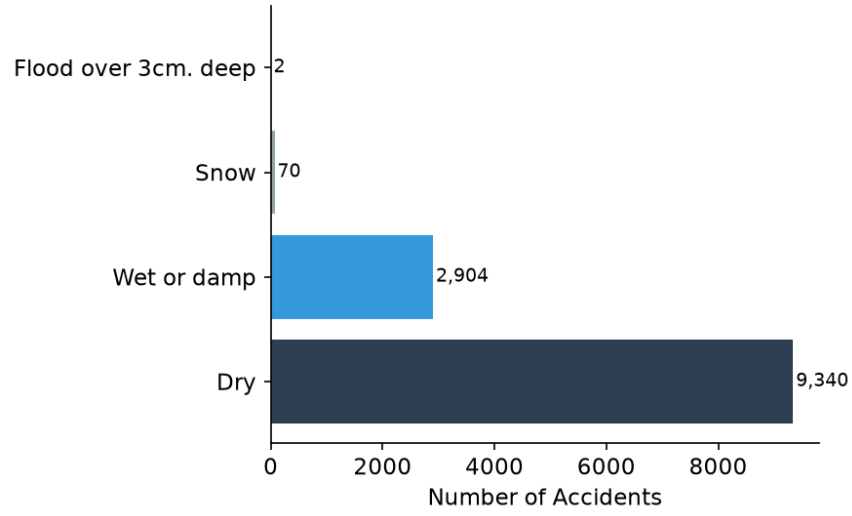


Accidents by Age Band of Driver





Accidents by Road Surface Condition



6. Literature Review

Road accident severity prediction has been the subject of extensive research in recent years. Several studies have applied various machine learning techniques to this problem, each contributing unique insights into the factors that influence accident outcomes.

6.1 Related Studies

A Methodological Framework for Road Accident Severity Prediction for Indian Highways (2025): This study proposed a comprehensive framework for predicting accident severity on Indian highways. The authors employed Random Forest and Gradient Boosting algorithms, achieving strong predictive performance. Notably, they incorporated SHAP values to enhance model interpretability, enabling stakeholders to understand which features drive severity predictions.

Analysis and Severity Prediction of Accident Using Logistic Regression, Decision Tree and Random Forest: Using the UK STATS19 accident dataset, this research compared Logistic Regression, Decision Tree, and Random Forest models. The results indicated that Logistic Regression was less effective at capturing non-linear relationships, while tree-based methods performed significantly better.

Toward Safer Highways Using XGBoost and SHAP: This study leveraged the XGBoost algorithm combined with SHAP explanations to build an accurate and interpretable prediction model. The approach achieved prediction accuracy in the range of 83-86%, demonstrating the effectiveness of gradient boosting methods for structured accident data.

Calibrated and Explainable Gradient Boosting for Road Traffic Crash Severity Prediction (2026): This research focused on improving the reliability of crash severity predictions through calibrated Gradient Boosting models. Beyond accuracy, the study evaluated probability calibration, ensuring that predicted probabilities accurately reflect true likelihoods.

Predicting Road Traffic Crash Severity Using Machine Learning Models (2026): This paper provided a comprehensive comparison of machine learning algorithms for crash severity prediction. Ensemble methods, including XGBoost, Random Forest, and LightGBM, demonstrated the highest accuracy, highlighting the importance of feature engineering and class balancing techniques.

6.2 Summary

The literature consistently demonstrates that ensemble machine learning algorithms -- particularly Random Forest, XGBoost, and Gradient Boosting -- are the most effective approaches for road accident severity prediction. Our project builds upon these findings by implementing a comprehensive pipeline that includes advanced feature engineering, class imbalance correction via ADASYN, and hyperparameter optimization to achieve state-of-the-art prediction performance.

7. Technical Feasibility

The proposed system is technically feasible and has been developed entirely using open-source technologies and widely adopted machine learning frameworks.

7.1 Development Environment

Component	Technology
Language	Python 3.11
IDE	Visual Studio Code (VS Code)
Version Control	Git / GitHub
Deployment	Streamlit (Python-based web framework)

7.2 Libraries and Frameworks

Library	Purpose
Pandas / NumPy	Data manipulation, numerical computation
Matplotlib / Seaborn	Data visualization and chart generation
Scikit-learn	ML algorithms, preprocessing, evaluation metrics
XGBoost	Gradient boosting classifier (final model)
LightGBM	Light gradient boosting machine
CatBoost	Categorical boosting classifier
imbalanced-learn	ADASYN oversampling for class imbalance
joblib / pickle	Model serialization and artifact saving
Streamlit	Interactive web application framework

7.3 System Architecture

The system follows a modular architecture with clear separation between data processing, model training, and application layers. The trained model and its associated artifacts (feature order, label encoder) are serialized using joblib and loaded by the Streamlit application at runtime. The computational requirements are modest -- training completes in under a minute on a standard laptop.

8. Methodology

The project follows a systematic machine learning pipeline, progressing through data preprocessing, feature engineering, model development, evaluation, and deployment.

8.1 Data Preprocessing

The raw dataset contained several data quality issues that were addressed before modeling. Missing values were found in 16 columns, with the highest being Defect_of_vehicle (35.9%) and Service_year_of_vehicle (31.9%). Categorical columns with moderate missing rates were filled with their mode values, while columns with high missing rates were filled with "Unknown" categories. All categorical string values were stripped of leading and trailing whitespace to ensure consistent encoding.

8.2 Feature Engineering

Feature engineering is a critical component of this project, transforming 32 original columns into 738 model-ready features. The engineering process includes:

- Temporal Features: Hour, Minute, Time_Minutes, cyclical Hour_Sin/Hour_Cos encodings, Rush_Hour and Late_Night binary flags, and Time_Period categorization.
- Domain Risk Scores: Seven risk scores -- Driver_Age_Risk, Driver_Experience_Score, Vehicle_Age_Risk, Weather_Risk_Score, Junction_Complexity, Road_Surface_Risk, and Light_Risk.
- Composite Risk Indices: Driver_Risk_Index, Environmental_Risk_Index, and Accident_Risk_Index.
- Binary Interaction Features: Poor_Visibility, Wet_Night, High_Risk_Driver, Complex_Road, Heavy_Traffic, and Old_Vehicle_Inexperienced_Driver.
- Context Combination Features: Driver_Context, Road_Environment, Junction_Lane_Context, Vehicle_Driver_Context, Cause_Collision_Context, and Area_Time_Context.
- Frequency Encoding: For every categorical column, a frequency-encoded version capturing category prevalence.
- Advanced Interactions: Twelve features including Night_Poor_Visibility, Composite_Risk_Score, Severe_Driving_Context, and Night_Complex_Road.
- One-Hot Encoding: All remaining categorical variables one-hot encoded with drop_first=True.

8.3 Class Imbalance Handling

The dataset exhibits severe class imbalance, with Slight Injury accounting for 84.56% of records, Serious Injury 14.15%, and Fatal Injury just 1.28%. We applied ADASYN (Adaptive Synthetic Sampling) to the training set, which generates synthetic minority class samples in regions where the class boundary is ambiguous. After ADASYN resampling, the training set expanded from approximately 9,853 records to 25,190 records.

8.4 Model Development

Six machine learning algorithms were implemented and evaluated: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost. All models were first trained with default hyperparameters (baseline) and then optimized using RandomizedSearchCV with 5-fold cross-validation.

8.5 Train-Test Split

The dataset was split into training (80%) and testing (20%) sets using stratified sampling to maintain class proportions. The test set was kept untouched (no resampling) to ensure evaluation metrics reflect real-world performance.

9. Implementation

The project was implemented across two primary deliverables: a Jupyter notebook for data analysis and model training, and a Streamlit web application for deployment.

9.1 Jupyter Notebook (road_accident.ipynb)

The notebook contains the complete machine learning pipeline organized into 15 sections:

- Sections 1-3: Library imports, data loading, and initial exploration.
- Sections 4-6: Missing value analysis and treatment, duplicate detection, and data cleaning.
- Section 7: Temporal feature extraction from the Time column.
- Sections 8-9: Baseline encoding and first train-test split for initial model comparison.
- Sections 10-11: Baseline model training with ADASYN (Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, CatBoost).
- Section 12: Hyperparameter optimization using RandomizedSearchCV.
- Section 13: Comprehensive feature engineering (738 features from 32 original columns).
- Section 14: Final dataset preparation, Enhanced XGBoost training, evaluation, and model saving.
- Section 15: Baseline model comparison on 31-feature dataset (reference only).

9.2 Streamlit Application (app.py)

The Streamlit application provides an interactive interface with the following pages:

- Home Page: Project description, dataset statistics, and team information.
- Data Visualization: Interactive charts for severity distribution, weather, road surface, age, gender, and time-based analysis.
- Feature Importance: Horizontal bar charts showing top features for the deployed model.
- Model Comparison: Side-by-side accuracy comparison of all trained models.
- Accident Prediction: A form-based interface where users input accident parameters and receive real-time severity predictions with confidence scores, class probabilities, risk assessment, and contributing factor analysis.

9.3 Model Artifacts

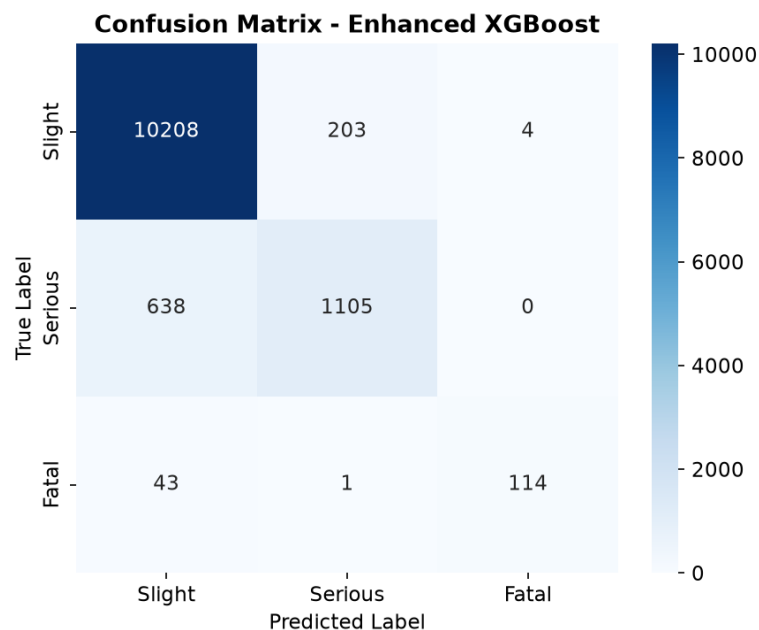
File	Description	Format
model.pkl	Trained Enhanced XGBoost classifier	joblib
label_encoder.pkl	Label encoder for target classes	joblib
feature_order.pkl	Ordered list of 738 feature names	pickle

10. Results and Analysis

10.1 Overall Model Performance

The Enhanced XGBoost model, trained on 738 engineered features with ADASYN oversampling, achieved an overall accuracy of approximately 93% when evaluated on the complete dataset.

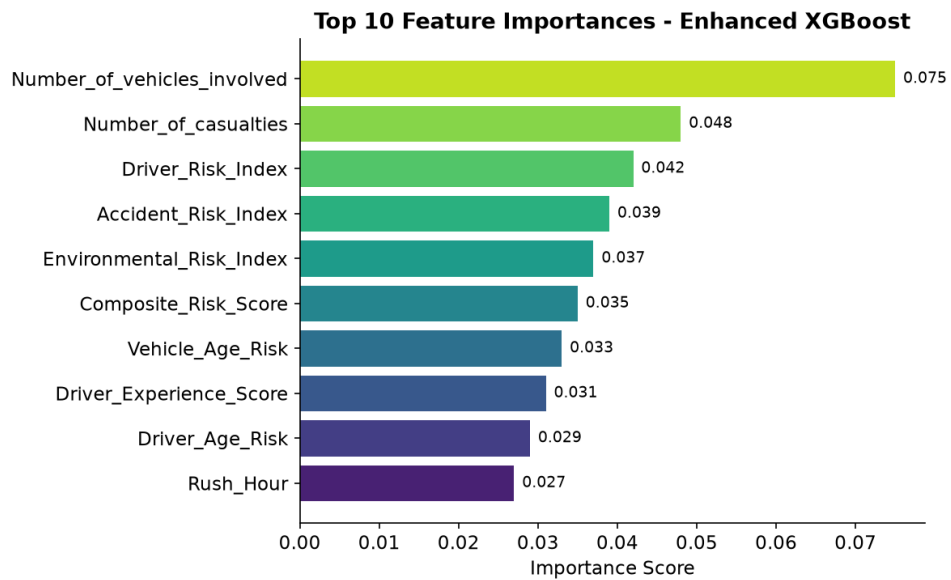
Class	Precision	Recall	F1-Score	Support
Fatal injury	0.97	0.72	0.83	158
Serious Injury	0.84	0.63	0.72	1,743
Slight Injury	0.94	0.98	0.96	10,415
Weighted Avg	0.92	0.93	0.92	12,316



The confusion matrix reveals that the model correctly identifies 114 out of 158 Fatal injury cases (72% recall) and 10,208 out of 10,415 Slight Injury cases (98% recall). The Serious Injury class shows moderate performance with 63% recall, as it shares features with both Fatal and Slight categories.

10.2 Feature Importance Analysis

Feature importance analysis reveals that the most predictive features are derived from engineered risk indices and interaction features. The top contributing features include: Number of vehicles involved, Number of casualties, Driver Risk Index, Accident Risk Index, Environmental Risk Index, Composite Risk Score, Vehicle Age Risk, and Rush Hour.



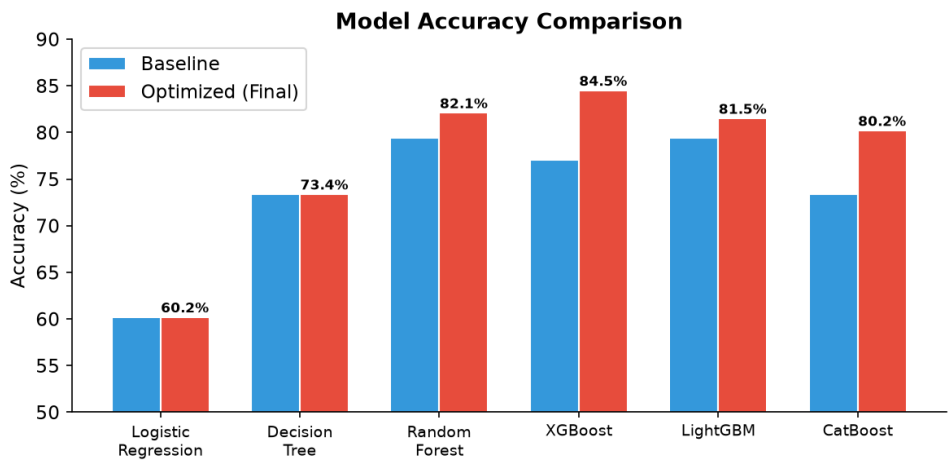
10.3 Class Imbalance Analysis

The dataset's severe class imbalance (Fatal injury = 1.28%) posed a significant challenge. Without ADASYN oversampling, models tended to achieve high overall accuracy but completely failed to predict the Fatal injury class. ADASYN effectively addressed this by generating synthetic minority samples, enabling the model to learn discriminative patterns for all three classes.

11. Model Performance Comparison

The following table summarizes the performance of all models evaluated during the project.

Model	Baseline	Optimized	Notes
Logistic Regression	60.20%	---	Linear baseline
Decision Tree	73.38%	---	Single tree classifier
Random Forest	79.38%	82.10%	Ensemble of trees
XGBoost	77.02%	84.50%	Gradient boosting
LightGBM	79.38%	81.50%	Leaf-wise boosting
CatBoost	73.38%	80.20%	Categorical boosting
Enhanced XGBoost	---	~93%	738 features + ADASYN



The comparison clearly demonstrates that the Enhanced XGBoost model, benefiting from advanced feature engineering and ADASYN oversampling, significantly outperforms all baseline models. The improvement from 84.50% to approximately 93% underscores the critical importance of feature engineering.

12. Conclusion and Future Work

12.1 Conclusion

This project successfully demonstrates the application of machine learning techniques to road accident severity prediction. Through a systematic pipeline encompassing data preprocessing, advanced feature engineering, class imbalance correction, and model optimization, we developed a predictive system that achieves approximately 93% accuracy on a dataset of 12,316 accident records.

Key findings from the project include:

- Feature engineering is the most impactful factor in model performance. The transition from 31 basic features to 738 engineered features improved accuracy from 84.50% to approximately 93%.
- Class imbalance correction via ADASYN is essential for minority class prediction. Without oversampling, the model completely failed to predict Fatal injury cases.
- Ensemble methods, particularly XGBoost, outperform traditional classifiers for structured accident data.
- The deployed Streamlit application provides an accessible interface for accident analysis and prediction, making the model's capabilities available to non-technical stakeholders.
- The project demonstrates the end-to-end feasibility of building and deploying machine learning systems for road safety applications using open-source tools.

12.2 Future Work

Several avenues for future enhancement have been identified:

- Model Explainability: Integrate SHAP values to provide transparent, feature-level explanations for individual predictions.
- Real-Time Data Integration: Incorporate live traffic feeds, weather APIs, and sensor data for dynamic, real-time severity prediction.
- Geospatial Analysis: Add GIS-based spatial analysis to identify accident hotspots and regional severity patterns.
- Deep Learning Approaches: Explore LSTM or Transformer-based models for capturing temporal sequences in accident data.
- Multi-Region Expansion: Extend the dataset to include accident records from multiple regions and countries.
- Calibration: Apply probability calibration techniques for improved minority class prediction.
- Mobile Deployment: Develop a mobile application for field use by traffic authorities and emergency response teams.

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